Udyam OneFlow · Solution architecture and data flows

Lead by Lord · VIT Chennai · Airtel SYNC 2026, Track A. Prototype built on the organisers' synthetic data.

Live in prototype · Simulated and labelled · Planned

Frontend · React, mobile-first

React mobile-first client
single-page web app

English · Hindi · Tamil
plus voice input

WhatsApp-style interface
simulated channel

Partner cockpit
ranked leads, routing, tracked links

1 HTTPS/JSON

6

Backend · FastAPI

AI advisor
profile extraction, plain-language reply

LLM adapter + output validator
language only; invented numbers rejected

3

Recommendation engine
eligibility, unlock paths, catalogue pricing

Local OCR + KYC validation
RapidOCR, up to 22 checks

Lead intelligence
propensity + value model, partner routing

Order management + tracker
5 stages, notifications (sim)

read catalogue 2

4 registry lookup, write fields

status + notifications

3

masked text only
no documents

External services

Gemini / OpenRouter
LLM, with fallback

4

DigiLocker
document fetch

6

WhatsApp Business API
messaging

5

Aadhaar OTP e-sign
signature

e-NACH mandate
payment mandate

OSS/BSS provisioning
activation hand-off

Data layer · SQLite + supplied data pack

Product	Lead	Mock GSTIN	Mock Udyam	Document store
catalogue (25)	dataset (5,000)	registry (3,534)	registry (2,723)	short retention

Data flows

1. Customer input and document upload (HTTPS/JSON)
2. Profile and catalogue retrieval
3. AI interpretation, then rules-based recommendation
4. KYC extraction and registry validation
5. Order creation and mandate simulation
6. Activation status and notifications

Synthetic data only. Rules decide every price; the LLM never sees documents.